

Supplementary Material: Failure-Aware Smartphone Inertial Dead Reckoning Under GNSS Outages

Article Title: Failure-Aware Smartphone Inertial Dead Reckoning Under GNSS Outages: An Empirical Study of Learned Velocity, Sensor Fusion, Map Constraints, and Distribution Shift

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Overview

This supplementary document provides extended tabular data, detailed per-trip forensic metrics, hardware profiling logs, and mathematical derivations supporting the main article. All evaluations utilize the public IO-VNBD benchmark dataset (Onyekpe et al., 2021) and on-device Android mobile execution on a Google Pixel 7a smartphone.

Supplementary Table S1: Per-Trip Performance Breakdown Under 60-Second GNSS Blackouts

The main manuscript presents aggregate and category-level metrics across the 13 usable held-out test routes. Table S1 provides the comprehensive route-by-route results for both unassisted strapdown inertial dead reckoning and the proposed NavSense adaptive fusion engine.

Table S1. Detailed route-by-route evaluation across all 13 usable held-out test trips under 60-second satellite outages.

Trip ID	Route Environment	Reference Distance (m)	Pure Strapdown Drift (m)	Pure Strapdown Drift (%)	NavSense Adaptive Drift (m)	NavSense Adaptive Drift (%)	SIH Benchmark Status (<10%)	Observed Dynamic Characteristics
Vta21	Suburban Town	786.7	406.8	51.45%	61.5	7.88%	MET (PASS)	Steady suburban cruising along arterial corridor; bounded drift
Vta22	Suburban Town	641.3	176.7	27.72%	107.7	16.84%	Unmet	Frequent stop-and-go queue; velocity dampens with traffic
Vta23	Suburban Town	570.1	352.9	62.40%	156.8	26.16%	Unmet	Sharp 90-degree intersection turn during blackout window
Vta24	Suburban Town	384.8	293.6	73.42%	57.0	14.65%	Unmet	Commercial zone corridor; low-speed transient manoeuvres
Vta26	Suburban Town	258.7	292.8	574.65%	66.3	47.31%	Unmet	Short transit crawl; large percentage due to short travel distance

Trip ID	Route Environment	Reference Distance (m)	Pure Strapdown Drift (m)	Pure Strapdown Drift (%)	NavSense Adaptive Drift (m)	NavSense Adaptive Drift (%)	SIH Benchmark Status (<10%)	Observed Dynamic Characteristics
Vta27	Suburban Town	818.9	306.5	37.18%	95.9	12.68%	Unmet	Signalised arterial road with moderate speed transitions
Vta28	Suburban Town	565.3	281.8	64.92%	120.6	25.30%	Unmet	Roundabout entry and deceleration into residential zone
Vw12	Mountain Route	1500.2	650.3	43.40%	45.3	3.01%	MET (PASS)	Continuous mountain descent; highly dynamic centripetal cues
Vw14a	Mountain Route	1511.1	312.6	21.06%	130.4	8.54%	MET (PASS)	Rural mountain corridor with sweeping curves and steady throttle
Vw14b	Mountain Route	1264.5	432.8	42.48%	121.0	10.19%	Unmet	Consecutive hairpins; centripetal acceleration transitions
Vw16a	Mountain Route	899.9	388.5	51.76%	136.1	16.88%	Unmet	High curvature hill climb with frequent throttle modulations
V-Vfa02	Motorway	1460.3	313.9	25.44%	157.2	11.69%	Unmet	Unaccelerated high-speed cruising; negative TCN regression bias
Vw15	Stationary Control	1.3	5.6	450.48%	0.82	63.83%	Unmet	Zero-velocity control trip; standstill clamped by causal ZVD
Mean / Total	All 13 Routes	820.2	324.21	89.66%	96.65	16.76%	3 / 13 (23.08%)	Complete Held-Out 60-s Evaluation Set

Note: In accordance with the experimental protocol, trips with total durations shorter than 70.0 seconds are excluded from the 60-second evaluation set because they cannot accommodate the 10-second pre-outage calibration window plus the 60-second blackout duration. Macro-average normalized drift weights each dynamic trip equally, excluding the stationary control trip Vw15 from the percentage mean per Equation (13) to prevent division-by-zero distortion.

Supplementary Table S2: Spectral Vibration Speed Audit Across 64 Vehicle Trips

To evaluate the hypothesis that vehicle road and engine vibration captured by consumer smartphone IMUs provides a reliable proxy for vehicle forward speed, Table S2 summarizes the spectral census across all 64 passenger-car trips in the IO-VNBD dataset (>450,000 one-second FFT windows).

Table S2. Cross-trip spectral vibration census across four standardized road categories.

Road Environment	Trips Analyzed (N)	Total 1-s Windows	Dominant Spectral Peak (Hz)	Pearson Correlation (r) with Speed	Spearman Correlation (ρ) with Speed	Physical Interpretation
Suburban Towns (Vta)	30	218450	2.24 ± 0.18 Hz	-0.028	-0.024	Consistent with low-frequency vehicle-body/chassis dynamics
Dense Urban (Vtb)	12	40890	2.31 ± 0.22 Hz	-0.041	-0.035	Consistent with road-surface excitation and low-frequency vehicle dynamics
Mountain Routes (Vw)	20	170920	2.45 ± 0.15 Hz	-0.019	-0.016	Consistent with chassis/body dynamics under curved-road motion
Motorways (V-fa)	2	39180	2.86 ± 0.12 Hz	-0.038	-0.031	Persistent spectral peak across speed bands; physical mechanism not directly identified
Complete Corpus	64	>450,000	2.2–2.5 Hz	-0.032	-0.028	No reliable speed signal identified at 10 Hz feature rate

Methodological Note: Power spectral density was computed causally using Welch periodograms across 1.0-second sliding windows (10 samples, zero-padded to 64 points) on tri-axial specific forces and angular velocities.

Supplementary Table S3: On-Device Android Execution Profiling and Numerical Parity

Table S3 details the execution latency, memory utilization, and cross-platform numerical parity between the mobile Android deployment (Java/EJML/ONNX Runtime) on a Google Pixel 7a and the 64-bit Python reference engine.

Table S3. Mobile execution benchmarks and cross-platform numerical parity on Google Pixel 7a.

Pipeline Subsystem	Implementation Language	Runtime Framework	Mean Execution Latency	Peak Latency (99th %tile)	Maximum Discrepancy vs. Python 64-bit Engine
IMU Ingestion & Decimation	Kotlin	Android SensorManager HAL	0.12 ms	0.35 ms	Identical (Zero difference)
Coordinate Leveling & PCA	Java	Native Linear Algebra	0.24 ms	0.58 ms	Leveling Angle: $<0.005^\circ$
TCN Forward Inference	C++ / Java Bindings	ONNX Runtime Mobile v1.16	5.82 ms	7.84 ms	Speed: <0.001 m/s
15-State ESKF Propagation	Java	Efficient Java Matrix Library (EJML)	0.94 ms	1.38 ms	Position: 0.24177 m; Heading: 0.01829°
Decoupled Velocity Damping	Java	Custom EJML Matrix Slice	0.18 ms	0.32 ms	Velocity: 0.02117 m/s
Open-Loop Vector Mapping	Kotlin	OSMDroid Offline Engine	1.85 ms	3.42 ms	Render: <1 frame

Pipeline Subsystem	Implementation Language	Runtime Framework	Mean Execution Latency	Peak Latency (99th %tile)	Maximum Discrepancy vs. Python 64-bit Engine
Total Navigation Epoch Loop	Mixed	End-to-End Android Pipeline	9.15 ms	13.89 ms	Within the 100-ms epoch budget

Supplementary Section S1: Mathematical Derivation of Causal Receptive Field Reach

The causal Temporal Convolutional Network (TCN) operates over a sliding temporal buffer of $W = 100$ samples (10.0 s at 10 Hz). To ensure that predictions depend strictly on past motion dynamics with zero dependency on future epochs, causal convolutions are enforced via asymmetric left-padding.

For a 1D convolutional network with L dilated residual blocks, kernel size k , and dilation factor d_l at layer l , the mathematical receptive field reach R (expressed in sample steps) is given by:

$$R = 1 + \sum_{l=0}^{L-1} (k - 1) \cdot d_l$$

In the TCN architecture, the parameters are:

- * Number of dilated residual layers: $L = 4$
- * Convolutional kernel size: $k = 3$
- * Dilation schedule: $d_l \in \{1, 2, 4, 8\}$ for $l \in \{0, 1, 2, 3\}$

Each residual block contains two sequential dilated convolutional layers with identical dilation d_l . Substituting these values:

$$R_{\text{block},l} = 2 \cdot (k - 1) \cdot d_l = 2 \cdot (3 - 1) \cdot d_l = 4d_l$$

Summing across all four dilated blocks:

$$R = 1 + \sum_{l=0}^3 4d_l = 1 + 4 \cdot (1 + 2 + 4 + 8) = 1 + 4 \cdot 15 = 61 \text{ samples}$$

At the nominal 10.0 Hz sampling rate, 61 samples correspond to an effective causal look-back interval of:

$$\tau_{\text{causal}} = (61 - 1) \times 0.10 \text{ s} = 6.0 \text{ seconds}$$

representing input dependency on the causal interval $[t - 60, t]$.